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Myofascial pain – A major player in musculoskeletal pain

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ABSTRACT

Myofascial pain is a soft tissue pain syndrome with local and referred musculoskeletal pain arising from trigger points. Myofascial pain and myofascial pain syndromes are among some of the most common acute and chronic pain conditions. Myofascial pain can exist independently of other pain generators or can coexist with or is secondary to other acute and chronic painful musculoskeletal conditions. Myofascial pain is most effectively treated with a multimodal treatment plan including injection therapy (known as trigger point injections, physical therapy, postural or ergonomic correction, and treatment of underlying musculoskeletal pain generators. The objectives of this review are to outline the prevalence of myofascial pain, describe the known pathophysiology of myofascial pain and trigger points, discuss the clinical presentation of myofascial pain, and present evidence-based best practices for pharmacologic, non-pharmacologic, and interventional treatments for myofascial pain.

1. Introduction

Myofascial pain is a soft tissue pain syndrome with local and referred musculoskeletal pain arising from trigger points (TrPs). Myofascial pain is a common cause of acute and chronic pain. Myofascial pain can exist independently of other pain generators, known as primary myofascial pain syndrome (MPS). Common primary MPS diagnoses include: piriformis syndrome, iliopsoas-related pain, and pain related to compression of the brachial plexus by the scalene muscles (neurogenic thoracic outlet syndrome). Frequently, myofascial pain coexists with or is secondary to other acute and chronic painful musculoskeletal conditions including (1) head and neck disorders (temporomandibular disorders, cervical degenerative disc disease, cervical facet arthropathy, neck pain after whiplash injury, cervicobrachial syndrome, and cervicogenic or chronic tension-type headache), (2) thoracolumbar back disorders (degenerative disc disease, kyphosis, scoliosis, lumbar facet arthropathy), (3) pelvic pain, and (4) upper and lower extremity pain disorders. Myofascial pain is most effectively treated with a multimodal treatment plan including injection therapy (known as trigger point injections (TPIs)), physical therapy, postural or ergonomic correction, and treatment of underlying musculoskeletal pain generators.

TrPs generating myofascial pain are localized painful areas of skeletal muscle containing taut bands that can be exquisitely sensitive to digital pressure. TrPs may be active or latent. Active TrPs are present in patients with painful regional conditions. Latent TrPs are asymptomatic but may be revealed by deep palpation on physical examination. Latent TrPs have been found in the shoulder girdle muscles of 45%–55% of healthy young adults [1].

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The objectives of this review are to outline the prevalence of myofascial pain, describe the known pathophysiology of myofascial pain and trigger points (TrPs), discuss the clinical presentation of myofascial pain, and present evidence-based best practices for pharmacologic, non-pharmacologic, and interventional treatments for myofascial pain.

2. Prevalence and epidemiology

The prevalence of myofascial pain can be difficult to determine due to the fact that there are no definitive diagnostic criteria that must be met by patients. The reported prevalence of myofascial pain has a large range with authors suggesting multiple different values. It can be noted that anywhere from 20 to 95% of patients who present to their general medicine physicians and pain management centers with some sort of musculoskeletal pain will be diagnosed with myofascial pain. The people most affected by this condition are between the ages of 27–50 years. It is also most commonly seen in patients with chronic tension-type headaches, temporomandibular disorders, pain in the face-jaw region, and in post-whiplash syndrome.

Some papers cite that the prevalence of myofascial pain is up to 85% in the general population with variable rates between males and females [2]. While other articles suggest that myofascial pain is more common among females [2]. It has also been estimated that around 85% of patients visiting chronic pain clinics and 30% of the patients visiting internal medical clinicals have myofascial pain. It has even been cited that myofascial pain is the most common cause of pain, responsible for 85% of back pain and 54.6% of chronic head and neck pain [3]. Another article cited that myofascial trigger points have a lifetime prevalence of up to 85% in the general population and can evoke a pattern of referred pain as well [4]. These statistics once again emphasize the number of patients presenting with this type of pain and demonstrates the importance of recognizing and treating these patients correctly.

One study investigated the prevalence of MPS in patients with non-specific neck pain that presented to their family medicine doctor and found that 100% of these patients were diagnosed with MPS. There have also been several studies in cancer patients looking at different causes of their pain and how to better treat this specific patient population. A specific study found that out of thirty-four patients with incurable cancer, that 29% and 50%, when determined by Rivers criteria and Simons criteria respectively, had MPS. MPS was also noted to be commonly diagnosed in patients receiving treatment of head and neck cancer [3]. Once again, these statistics point to the importance of myofascial pain. It can present in many different ways but is most often described as a chronic condition. It can present acute but is the leading cause of chronic and persistent regional pain, including shoulder pain, chronic back pain, tension type headaches, and facial pain [2].

A systematic literature review was performed to evaluate the prevalence of myofascial trigger points in patients with neck and back pain [4]. They reviewed fourteen articles and found that within spinal pain, patients with neck pain were found to have the highest prevalence of myofascial trigger points, with the most common locations being the trapezius descendens, levator scapulae, and suboccipitales muscles. It was noted in this study that further investigation is needed to determine the prevalence of myofascial trigger points in patients with low back pain as research in this area is lacking.

A study performed by Skootsky and colleagues looked at the prevalence of MPS in patients presenting to primary care general medicine clinic in the academic setting [5]. Patients were given a questionnaire that asked about pain and to mark its location on a body diagram. Patients with pain were then examined for discrete tender points to see if a clinical diagnosis of myofascial pain could be determined. A total of 201 patients were asked to participate and after some excluded, the study was left with 172 participates. Of the 172, 54 (31%) had initial complaints of musculoskeletal pain. Of these patients, 16 met criteria of MPS, representing 30% of patients with musculoskeletal pain and 8% of the 201 originally screened.

Another study done by Ishiki and colleagues looked at MPS and patients with incurable cancer. In this study, 34 patients with incurable cancer were identified. It was noted that as many as 40% of these patients had MPS. This information can be used moving forward to help treat these patients and avoid medications that may not be effective [6].

3. Clinical presentation

Myofascial pain and MPS may occur after injury, chronic strain with repetitive microtrauma, or without clear precipitating event. Aberrant body mechanics or postural abnormality may initiate or perpetuate the problem. Subjectively, patients may report symptoms such as regional body pain, referred dull, deep, aching pain in a specific anatomic region, limited range of motion of the affected areas, impaired muscle coordination, muscle fatigue, and weakness [7].

4. Diagnosis

As with other pain disorders, diagnosis of myofascial pain requires a careful history and physical exam for correct diagnosis. A thorough musculoskeletal exam can help differentiate between myofascial pain and other causes of pain as well as identification of orthopedic or neurologic factors that may contribute to the patient's symptoms or be a cause of secondary MPS. The hallmarks of MPS are myofascial trigger points (TrPs) or localized areas of skeletal muscle with palpable, rope-like taut bands, hypersensitive spots, and referred pain, with at least two of three needed for diagnosis generally [8]. Successful relief of pain with trigger point injections can help confirm diagnosis. It's important to distinguish between local tenderness and hyperalgesia with referred pain as the former is typical of tender spots while the latter is associated with trigger points [9]. The ability to palpate these taut bands is dependent on the examiner's training and skill and depth and size of the muscle [7]. TrPs can be exquisitely sensitive to digital pressure, so much so that palpation can elicit a "jump sign" [7], or involuntary muscle contraction. In some cases, patients can also have associated autonomic phenomena including flushing, hyperhyrosis, temperature changes, piloerection, and dizziness [10,11]. Two distinct clinical

categories have been used to classify TrPs - latent and active. Latent TrPs do not cause spontaneous pain but rather with application of pressure to the TrP, producing referred pain while active TrPs may cause pain at rest, weakens the muscle, prevents full range of motion, and generally produce larger pain patterns than latent [11].

While a general consensus has been made in regard to diagnostic criteria for TrPs, reliability between examiners has historically been inconsistent [12]. There were high levels of variability depending on examiner skill level, examination technique utilized, variations in muscle anatomy, and location of the taut bands (i.e. deep or superficial muscles). More objective tools are being developed to help reliably identify TrPs and guide treatment. Pressure algometer measurements can be used to measure pressure pain thresholds for a given muscle group to help quantify sensitivity to pressure, or the pressure threshold measurement (PTM). PTMs can aid in the diagnosis of TrPs as well as provide longitudinal data regarding the efficacy of interventions, although changes in patient's levels of tension and anxiety can increase variability between visits [13]. Additional methods such as ultrasound, vibration sonoelastography (VSE), muscle pain detection devices (MPDD), Doppler flow studies, and magnetic resonance elastography (MRE) have been studied as advanced tools for objective diagnosis, but further studies are needed to determine which tool is most effective [11].

Distinct patterns of TrP findings may reveal itself after a given insult. Common TrP locations associated with myofascial pain syndrome can be categorized by the anatomical region of the muscle group or the location of the referred pain. The common regions for myofascial pain syndrome include cervical, lumbar, abdominal, and pelvic, which will be further discussed below.

Differential diagnosis of MPS should include (1) musculoskeletal and neuropathic disorders such as arthritis, degenerative disk disease, radiculopathy, bursitis, and tendonitis; (2) autoimmune or infectious etiologies; (3) metabolic and endocrine dysfunction including hypothyroidism; (4) psychiatric disorders including depression and anxiety; and (5) fibromyalgia.

Some have postulated MPS as a precursor to fibromyalgia, but there's a host of contradictory evidence against this theory. While MPS and fibromyalgia share some commonalities, they are two separate entities and several well-documented findings argue against the causal relationship between MPS and fibromyalgia (Table 1). Patients with MPS may present similarly to patients with fibromyalgia, but the key distinguishing factor of MPS is localized TrPs rather than widespread tender spots. Additionally, fibromyalgia pain does not elicit patterns of referred pain when palpated unlike TrPs. TrPs can still be present in patients with fibromyalgia and patients may still find treatment of TrPs beneficial in reducing overall pain.

5. Regional myofascial pain syndromes

Common muscle groups associated with myofascial pain syndrome can be categorized by the regions in which the muscles anatomically exist or where the muscles can refer pain to. The common regions for myofascial pain syndrome include: cervical, shoulder, chest, abdomen, low back, pelvis, hip, knee, and ankle/foot. Box 1 describes the muscles that can contribute to pain in each of these regions. In this section, we will also discuss the most common regional MP syndromes including cervical MP, lumbar MP, abdominal MP, pelvic MP, and piriformis syndrome.

5.1. Cervical myofascial pain

Cervical TrPs can be found in the posterior cervical muscles (i.e. splenius capitus and cervicis, semispinalis capitus and cervicis), sternocleidomastoid, upper trapezius, levator scapulae, and facial muscles such as the temporalis and masseter muscles, with each muscle group having a unique pain referral pattern. For example, masseter TrPs may cause referred pain to the jaw or posterior occiput whereas splenius capitis TrPs may refer pain to the top of the head. The four most common muscles to develop TrPs and cause headaches are the temporalis, upper trapezius, splenius capitus, and the sternocleidomastoid [7]. Cervical TrPs are frequently associated with autonomic dysfunction, including sweating, lacrimation, flushing, and temperature changes, as well as neuro-otologic symptoms such as dizziness and tinnitus [7]. Subjectively, patients may report paresthesias, impaired muscle coordination, temporomandibular symptoms, weakness, blurred vision, or twitches [14].

Table 1
Clinical distinctions between myofascial pain syndrome and fibromyalgia syndrome.

Clinical Feature	Myofascial Pain	Fibromyalgia
Pain pattern	Local or regional	Generalized
Least distribution	A single muscle	Widespread hyperalgesia and allodynia to palpation
Muscle spasm	+++	++
Trigger points	Local, regional	Not a feature
Tender points	Not a feature	Common, widespread
Taut band	++	–
Twitch response	++	–
Referred pain	+++	–
Fatigue	+	++++
Sleep disturbance	+++	++++
Paresthesias	Regional	Distal
Headaches	Referred head pain	Associated with migraine headaches, tension type headaches
Irritable bowel	Not a feature	+++
Pathophysiology	TrPs	Central sensitization

Box 1
Muscles involved in Regional Myofascial Pain Syndromes

Head and Neck.

- Posterior cervical
  - •Splenius capitus and cervicis, semispinalis capitus and cervicis
- Sternocleidomastoid
- Upper trapezius
- Levator scapulae
- Facial muscles
  - •Temporalis
  - •Masseter

Shoulder.

- Trapezius
- Supraspinatus
- Infraspinatus
- Subscapularis
- Teres major and minor
- Latissimus dorsi
- Levator scapulae
- Posterior serratus
- Deltoid
- Pectoralis major and minor

Chest.

- Pectoralis major and minor
- Abdominal obliques
- Intercostal muscles
- Serratus anterior

Abdomen.

- Serratus anterior
- Abdominal obliques
- Rectus abdominus

Low Back.

- Psoas
- Quadratus lumborum
- Paraspinal muscles
  - •Iliocostalis
  - •Longimus thoracis
  - •Multifidi
- Abdominal obliques

Pelvis/Hip.

- Abdominal muscles
- Psoas
- Quadratus lumborum
- Gluteal muscles
- Piriformis
- Pectineus
- Semitendinosis
- Obturators
- Gemellae

Knee.

- Vastus medius and lateralis
- Hamstrings
- Gastrocnemius

Ankle/Foot.

- Gastrocnemius
- Soleus
- Tibialis anterior
- Fibularis longus and brevis
- Long flexor and extensor muscles of the leg
- Intrinsic foot muscles

5.2. Lumbar myofascial pain syndrome

The lumbar region contains many tissues capable of generating nociceptive input, with lumbar TrPs serving as an important contributor to chronic low back pain [15]. Chronic stress of hip flexor and hamstring muscles can result in pelvic tilt and worsening lumbar lordosis, increasing the risk of lumbar TrPs [16]. Examination of posture, spinal alignment, and gait disturbances can be beneficial in elucidating the cause of the low back pain. Additionally direct physical examination including musculoskeletal palpation can help identify underlying taut bands and TrPs. The two most commonly affected muscles resulting in low back pain are the quadratus lumborum and gluteus medius, with the multifidus, logissimus thoracis, and iliocostalis lumborum muscles as other common contributors.

5.3. Abdominal myofascial pain

Abdominal MPS, or chronic abdominal wall pain (CWAP) caused by trigger points, should be on the differential for chronic abdominal pain, especially in the setting of negative diagnostic tests such as labs and imaging. Abdominal TrPs are an important cause of refractory abdominal pain, causing constant, dull abdominal pain with intermittent flare-ups of referred flank, back, and groin pain. Unlike TrPs in other areas of the body, abdominal TrPs can develop not only from trauma to the area (i.e. physical, surgical) but also visceral inflammation, known as viscero-somatic convergence (VSC). It's important to note that while TrPs can develop via VSC, treating the underlying inflammation helps to prevent formation of new TrPs but is less beneficial in managing pain symptoms of established TrPs [17]. To help distinguish between parietal and visceral pain, the examiner can palpate the abdomen while the patient flexes the abdominal muscles. If there is increased pain with this maneuver, it's known as a positive Carnett's sign, indicating the pain is likely stemming from the abdominal wall rather than the underlying viscera [13]. This finding, along with other findings of TrPs (e.g. referred pain, "jump sign"), makes the diagnosis of abdominal TrPs more likely.

5.4. Pelvic floor myofascial pain

Pelvic floor myofascial TrPs are considered one of the most common diagnoses in patients with chronic pelvic pain, yet are frequently overlooked. MPS is associated with urinary, bowel, and sexual dysfunction and therefore is associated with other conditions such as vaginismus, interstitial cystitis, chronic constipation, and endometriosis [13]. Although pelvic TrPs can present similarly to primary dysmenorrhea, as pelvic pain in the absence of underlying pelvic organ pathology, pelvic MPS can have additional symptoms including autonomic symptoms and emotinal changes as well as physical exam findings consistent with TrPs [13]. While there are many ways to evaluate for pelvic TrPs, there is a lack of consensus over which test is superior [18]. Some studies have demonstrated the utility of single digit valginal palpation of the pelvic floor muscles for diagnosis [13]. Pain resulting from palpation of the TrPs can be referred to a variety of locations including the abdominal wall, gluteal, or quadratus lumborum muscles, with palpation of the TrPs able to reproduce the patient's pain, similar to their menstrual pain.

In regard to urinary dysfunction, studies have demonstrated a link between pelvic floor muscle TrPs and interstitial cystitis. Estimates of the prevelance of pelvic TrPs in patients with interstitial cystitis range from 50% to 90%, with the most commonly affected muscle being the levator ani [19]. Further research is being done to create an objective measure of pelvic floor muscle dysfunction, including the use of surface electromyography to aid in diagnosis [20].

5.5. Piriformis syndrome

Piriformis syndrome consists of pain in the buttock with or without radiation in the distribution of the ipsilateral sciatic nerve. It is considered the principal pain generator in approximately 8% of patients presenting with the buttock as the origin of pain. The syndrome can be a consequence of an abnormal relationship between the sciatic nerve and the piriformis muscle that results in irritation of

the sciatic nerve. A hypertrophic muscle, infection, or invasion of the muscle by tumor can cause pressure or irritation on the nerve [21, 22]. In 78%–84% of the population, the sciatic nerve passes in front of the muscle. In 12%–21% of individuals, the divided nerve passes through or posterior to the piriformis and is exposed to muscle contractions, which trigger sciatic symptoms [21].

There is no consensus set of diagnostic criteria and is often diagnosed from a history indicative of this condition with confirmatory physical exam with an often normal electrodiagnostic testing and imaging [23]. The syndrome should be considered in patients who have buttock pain (with or without sciatic symptoms), tenderness to palpation over the piriformis muscle, and a positive response to provocative maneuvers, including the following: Specific confirmatory physical exam testing for piriformis syndrome include the active piriformis test, the Beatty test, the FAIR test, and the Pace test [23].

6. Trigger point pathophysiology

Though common and well characterized by the presence of taut bands, muscle stiffness, and pain, the exact pathophysiology and cause of myofascial pain syndromes remains unknown [24]. Several theories have been postulated to explain this condition. It has been hypothesized that repetitive muscular strain results in increased oxidative metabolism and energy depletion in muscles resulting in increased muscle nociception. This increased nociception is believed to play a role in myofascial pain syndromes [25]. This increased nociception is particularly prevalent in type I muscle fibers with type III and IV muscle fibers mediating the pain and tenderness. Fiber transformation can also occur between the various muscle fiber types and it is possible that changes in neural patterns, mechanical loads, and hormonal factors can influence the variable development of muscle types [26]. These alterations can result in tissue injury which results in peripheral sensitization [27]. At the same time, to reduce myofascial pain, muscular hyperactivity and bracing occurs which can also result in peripheral sensitization [25].

Type I muscle fibers are slow twitch fibers with greater resistance to fatigue meanwhile fast twitch fibers are typically type II which are less fatigue resistant [28]. These slow twitch muscles are important to maintain static postures. Changes in expression of type I and type II fibers have been suggested as causes of myofascial pain. It has been found that in patients undergoing neck surgery for neck pain, biopsies of ventral neck muscles at the time of surgery had a high proportion of Type II-C fibers specific to transformational muscle development than Type I muscle fibers [29]. These findings suggest that transformational changes from type I to type II-C fibers may be associated with the development of pain as these were similar findings in neck muscle biopsies of patients with chronic musculoskeletal pain conditions such as rheumatoid arthritis [29,30]. Other studies have found that increased static postural muscle activation increases type I fiber expression while decreasing type II expression in patients with muscle pain. Interestingly, with age, the muscle density remains the same but there is a higher proportion of Type I muscle fibers [31]. This strain is thought to result in muscle unit hyperpolarization and decrease in local blood flow contributing to the development of myofascial pain [30].

Myofascial pain can be further classified as primary and secondary pain syndromes. Primary myofascial pain is a result of direct stimuli of macro or micro trauma including overload, postural changes, and repetitive motions which causes of initial insult [32]. These are typically the most common cause of myofascial pain syndromes. Secondary myofascial pain is a result of referred pain from distant structures. No formalized theory is available to explain secondary myofascial pain but theories have been postulated associating this condition with central sensitization and neurogenic inflammation which results in downstream nociceptive activation distant from the initial site of injury. Further, referred pain may be a resultant from a triggered allogenic focus where by noxious stimuli causes hyperexcitability at the dorsal horn where nerves from distal structures converge resulting in pain distal from the initial site of stimulation [32].

In a review article by Borg-Stein and Simmons, it is outlined that these macro and micro traumas ultimately result in abnormally functioning motor end plates which is the primary site of dysfunction in myofascial pain. This allows for an irregular build up and release of acetylcholine within the muscle fibers [33]. This irregular build up and release causes a sustained sarcomere contracture that results in increased muscle metabolism with local ischemia. This contracture creates the characteristic “trigger point”. Prolonged contractures ultimately creates an energy crisis that results in release of sensitizing substances that perpetuate the abnormal acetylcholine build and release. A byproduct of the sensitizing substances such as prostaglandin, bradykinin, capsaicin, serotonin, and histamine is the development of myofascial pain, first proposed by Simons et al. [33] This results in a cycle of hypoxia with subsequent release of vasoactive substances which locally sensitizes the nociceptors in the muscle [32]. EMG studies have been performed and corroborated these findings as trigger points have increased endplate noise though the diagnostic value and meaning of this finding has been disputed as it can occur with various other causes.

Aside from peripheral sensitization, central sensitization can also play a large role in the development of myofascial pain syndromes. There is significant overlap between the fibromyalgia and myofascial pain though the conditions themselves are quite different. Both involve chronic widespread musculoskeletal pain. However, fibromyalgia is thought to be associated with nervous system hyperexcitability and alterations in pain processing [34]. Meanwhile, myofascial pain is thought to be primarily peripherally mediated by these “taut bands” or “trigger points.” It is commonly accepted that these “trigger points” can contribute to the widespread pain peripherally in patients with centrally mediated pain conditions [34]. However, pain is complex and central process such as stress and anxiety can affect myofascial pain syndromes as well [30]. Prolonged presence of these “trigger points” can contribute to referral pain patterns beyond the nociceptive site, mediated by the spinal cord. This results in spinal neuroplastic changes in the dorsal horn, creating hyperexcitability of the nociceptor pathway [33].

6.1. Concurrent phenomenon of myofascial pain with other pain disorders

In clinical practice, it is not uncommon to encounter patients with myofascial pain along with other pain disorders, such as spinal

pathology, creating a concurrent phenomenon that poses diagnostic and therapeutic challenges. This is especially important when deciding on treatment options as the treatment of these concurrent issues can lead to significantly different procedural options with varying degrees of invasiveness.

6.2. Myofascial pain and spinal pathology

One of the most prevalent conditions associated with myofascial pain is spinal pathology, including degenerative disc disease, spinal stenosis, and disc herniation. The relationship between myofascial pain and spinal pathology is multifaceted. Spinal pathology can lead to altered biomechanics, resulting in muscle imbalance, postural changes, and increased stress on muscles, which can predispose individuals to developing myofascial trigger points. Conversely, myofascial trigger points can contribute to spinal pain by causing muscle dysfunction, altered movement patterns, and referred pain to adjacent spinal structures.

Several studies have investigated the concurrent occurrence of myofascial pain and spinal pathology. In a study by Gerwin et al. [35], the authors found that patients with chronic neck pain and cervical disc disease had a significantly higher prevalence of myofascial trigger points compared to those without disc disease. Similarly, a study by Chiarotto et al. [36]. reported a high prevalence of myofascial trigger points in patients with lumbar spinal stenosis, suggesting a potential link between spinal pathology and myofascial pain. There is also an element of neurologic factors that can arise from structural pathologies including radiculopathy that can create a degree of autonomic dysregulation and ultimately central spinal cord sensitization which can amplify the experience of myofascial pain.

6.3. Myofascial pain and other pain disorders

In addition to spinal pathology, myofascial pain can coexist with other pain disorders, such as fibromyalgia, temporomandibular disorders (TMD), and chronic tension-type headache (CTTH). Fibromyalgia is a chronic pain condition characterized by widespread musculoskeletal pain, fatigue, and tender points. Myofascial trigger points have been identified in a significant proportion of patients

Table 2
Summary of physical modalities for the treatment of myofascial pain syndrome. (Created by Tieppo Francio, V).

Physical Modality	Characteristic	Proposed Mechanism of Action
Muscle Stretching	Controlled elongation of muscles	Increases flexibility and range of motion Reduces muscle stiffness and enhances blood circulation
Postural Exercises	Targeted exercises to improve body alignment and posture	Corrects muscle imbalances Reduces strain on affected muscles and joints
Strengthening Exercises	Resistance training to build muscle strength	Improves muscle support around affected areas Enhances stability and reduces the risk of future injuries
Aerobic Exercises Cardiovascular Fitness	Moderate-intensity exercises to improve cardiovascular health	Increases blood flow and oxygen delivery to muscles Enhances overall fitness and promotes pain management
Neuromuscular reeducation	Techniques to improve the coordination of muscles and enhance proprioception to aid on pain reduction and resolution of triggers	Optimizes proprioception of the neuromusculoskeletal kinetic chain Improved motor control and muscle coordination
Heat Therapy	Application of heat to affected areas	Improves blood flow and tissue elasticity Relaxes muscles and reduces pain sensitivity
Cryotherapy	Application of cold to reduce inflammation	Reduces swelling and numbs the affected area Slows nerve impulses, providing pain relief
Transcutaneous Electrical Nerve Stimulation (TENS)	Application of low voltage electrical currents to the skin	Blocks pain signals to the brain Promotes the release of endorphins, natural painkillers
Therapeutic Ultrasound	Application of high-frequency sound waves	Promotes tissue healing through increased blood flow Relieves pain and inflammation through thermal effects
Massage Therapy	Manual manipulation of soft tissues	Breaks down adhesions and knots in muscles and improves blood flow and reduces muscle tension
Myofascial Release/Active Release	Gentle manipulation of the fascia and connective tissue	Releases tension in the fascia Improves flexibility and restores natural movement patterns
Low-level Laser Therapy	Application of low-level laser light to affected areas	Stimulates cellular activity and promotes tissue repair Reduces inflammation and relieves pain
Biofeedback	Use of electronic monitoring to provide real-time feedback on muscles	Increases awareness of muscle tension and facilitates control of muscles
Cognitive Behavioral Therapy (CBT)	Psychotherapy technique that facilitates behavioral change by altering patient's belief or thought patterns	Increases self-consciousness of pain Decreases trigger of pain
Extracorporeal Shock Therapy	Application of shock waves to the affected area	Stimulates healing processes and tissue regeneration Modulates pain signals and reduces inflammation
Acupuncture	Insertion of thin needles at specific points on the body following traditional acupuncture meridians	Stimulates nerve fibers and releases endorphins Modulates pain signals and promotes overall well-being
Dry Needling	Insertion of thin needles into trigger points	Stimulates trigger points, releasing muscle tension Modulates pain signals and promotes healing

with fibromyalgia, suggesting a possible overlap in the pathophysiology of these conditions [34].

Similarly, TMD is a common condition characterized by pain and dysfunction in the temporomandibular joint (TMJ) and surrounding muscles. Myofascial trigger points in the masticatory muscles are frequently observed in patients with TMD, and their presence has been associated with increased pain severity and reduced jaw function [25].

CTTH is another pain disorder characterized by recurrent episodes of headache with associated muscle tenderness. Myofascial trigger points in the cervical and shoulder muscles are commonly found in patients with CTTH, and their presence may contribute to the persistence and recurrence of headaches [37].

6.4. Risk factors for development of myofascial pain

Several risk factors have been identified for the development of myofascial pain, including age, gender, occupational factors, and psychological factors. Aging is associated with an increased prevalence of myofascial trigger points, possibly due to changes in muscle structure and function. Women are more likely to develop myofascial pain compared to men, which may be related to hormonal factors and differences in muscle physiology [32].

Occupational factors, such as repetitive or prolonged activities, poor ergonomics, and high levels of stress, can increase the risk of developing myofascial pain. Individuals who perform repetitive tasks or maintain prolonged static postures are more susceptible to developing myofascial trigger points due to muscle overuse and fatigue [38].

Psychological factors, including stress, anxiety, and depression, have also been implicated in the development of myofascial pain. Chronic stress can lead to muscle tension and the formation of trigger points, while psychological distress can lower the pain threshold, increasing the sensitivity to pain [32].

In conclusion, myofascial pain often coexists with other pain disorders, such as spinal pathology, fibromyalgia, TMD, and CTTH, creating a concurrent phenomenon that poses diagnostic and therapeutic challenges. Understanding the relationship between myofascial pain and other pain disorders is crucial for effective management and improving patient outcomes. Future research is needed to elucidate the underlying mechanisms and optimal treatment strategies for these complex pain conditions.

7. Treatment

A multitude of treatment options encompassing physical modalities, psychological approaches, oral and topical medications, injection therapy are available for managing myofascial pain syndrome. Treatment effectiveness may depend on the region of the myofascial pain syndrome. While an array of interventions exists, a lack of definitive consensus surrounds these modalities. Nevertheless, treatment endeavors should be oriented towards mitigating and rectifying perpetuating factors that have the potential to elicit myofascial pain and directed to patient education [32,39].

7.1. Physical modalities

Physical therapeutic modalities encompass a spectrum of approaches, including manual therapy techniques, exercises, stretching, physical therapy and physical modalities. Table 2 summarizes examples of physical therapy and modalities in the treatment of myofascial pain syndrome.

Management of myofascial pain syndrome is directed at deactivating trigger points and taut fibers and preventing recurrence. This can be achieved with physical modalities to induce a muscle stretch phenomenon, such as spray and stretch, post isometric relaxation technique, and local tissue stretch with pressure release, injection therapy and pharmacological methods [32]. Exercise constitutes a pivotal facet in the rehabilitation and management of musculoskeletal pain, playing a central role in enhancing flexibility, augmenting functional status, optimizing mood, and mitigating pain. The effectiveness of exercise therapy in myofascial pain syndrome is well-known. A gradual, progressive and comprehensive exercise program including stretching, strengthening and aerobic activity has physical and cognitive benefits [40]. While no specific physical modality or manual therapy technique has demonstrated unequivocal superiority, according to a recent systematic review and meta-analysis, myofascial release, muscle energy technique are frequently employed, and laser therapy and extracorporeal shockwave therapy may offer transient relief over placebo [41].

Table 3
Summary of pharmacological agents for myofascial pain syndrome (Created by Tieppo Francio, V).

Drug Class	Examples	Side Effects	Special Considerations
Nonsteroidal Anti-inflammatory Drugs (NSAIDs)	Ibuprofen, Naproxen, Meloxicam, Celebrex	Gastrointestinal upset, increased risk of bleeding, renal impairment	Preferred when associated symptoms of inflammation, acute on chronic
Muscle Relaxants	Cyclobenzaprine, Tizanidine, Baclofen	Drowsiness, fatigue, dizziness	Preferred when associated symptoms of sleep disturbance, chronicity
Tricyclic Antidepressants (TCAs)	Amitriptyline, Nortriptyline	Sedation, dry mouth, constipation	Preferred when associated symptoms of mood and sleep disturbances
Topical Analgesics (cream/patch)	Lidocaine, Diclofenac, Capsaicin	Allergic reaction, skin irritation	Preferred when associated symptoms of hyperalgesia
Serotonin/Norepinephrine Reuptake Inhibitors (SSRI/SNRIS)	Duloxetine, Venlafaxine	Nausea, insomnia, increased blood pressure	Preferred when associated symptoms of mood disturbance and hyperalgesia

Transcutaneous electrical nerve stimulation (TENS) has a short-term but not long-term effect on pain control and therapeutic ultrasound is commonly used, however the evidence of its beneficial effect remains inconclusive [42]. Traditional acupuncture has been shown to be beneficial, especially as adjunct to a multimodal approach, however a few studies have demonstrated superiority of acupuncture over pharmacotherapy [43–45]. Moderate evidence indicates that the application of dry needling on trigger points located in the lower back region of individuals with myofascial pain syndrome, particularly when combined with other therapeutic modalities, may be deemed advisable for alleviating pain intensity. Nevertheless, the clinical superiority of dry needling in enhancing functional disability and elucidating its sustained effects remains equivocal [46].

Muscle stretching, postural exercises, strengthening exercises, and cardiovascular fitness are some of the most useful rehabilitation strategies in myofascial pain syndrome [3]. Joint manipulation and strain counterstain techniques has been observed to be effective in myofascial pain, however there are no high-quality clinical studies to suggest long-term efficacy and most data is from by low quality evidence [47]. All non-pharmacological measures that reduce stress such as yoga, meditation, and cognitive behavioral therapy help may help reduce muscle tone, promoting an increase in the pain threshold and can benefit myofascial pain syndrome [48]. Patient education is essential, and the clinician should focus on building good rapport with the patient, validating their concerns, providing reassurance, empathy and understanding. Patients should be educated on underlying psychosocial factors, the importance of cardiovascular fitness, behavioral modifications and trigger avoidance [49].

7.2. Pharmacotherapy

In light of the limited understanding of the pathophysiology characterizing myofascial pain syndrome, the available repertoire of pharmacological interventions is notably constrained, predominantly targeting local musculature, peripheral nerves, and elements of the central nervous system. The pharmacotherapeutic strategies employed in the management of myofascial pain syndrome operate through mechanisms that may directly or indirectly impact the nervous systems. Consequently, a paramount consideration in the administration of these medications is the comprehensive evaluation of their safety profiles. Table 3 summarize the most common drug classes and side effects for the treatment of myofascial pain syndrome.

Nonsteroidal anti-inflammatory drugs (NSAIDs) represent the predominant pharmacological class employed in the management of myofascial pain syndrome. This preference can be attributed, in part, to their relatively modest adverse effect profile and widespread accessibility [50]. While NSAIDs and muscle relaxants are frequently prescribed, the present body of evidence regarding their efficacy is inconclusive [49]. Topical NSAIDs have been shown to be useful and they have fewer systemic side effects than oral NSAIDs but are often more expensive.

Topical agents, such as lidocaine have been shown to be effective in the management of myofascial pain syndrome, in comparison to placebo and to trigger point injection [51]. The use of local anesthetic creams or patches with different anesthetics (lidocaine or diclofenac sodium) has been explored, in addition to other alternatives such as capsaicin cream and studies have demonstrated superiority over a placebo patch [3].

Notwithstanding their prevalent utilization in clinical settings, muscle relaxants, exemplified by cyclobenzaprine or tizanidine, are subject to an empirical landscape wherein a majority of studies posit insufficient evidence to substantiate their efficacy in the context of myofascial pain syndrome, however these have good tolerability [Borg-Stein 2014]. Additional pharmacotherapeutic agents such as tricyclic antidepressants, serotonin and norepinephrine reuptake inhibitors may be utilized, in particular with the concomitant presence of psychiatric or mood disorders. Amitriptyline and Nortriptyline may be preferably used when depression, anxiety, insomnia are present. Benzodiazepines have numerous side effects and are traditionally not indicated and there are no published studies to support the use of opioids in myofascial pain [49,52].

7.3. Trigger point injections

Trigger Point Injection (TPI) stands as a widely utilized intervention for the management of myofascial pain syndrome. This procedure involves the guided insertion of a needle into a trigger point, with the primary objective of alleviating trigger point pain that has proven refractory to conventional physical modalities and therapeutic exercises [1]. TPI are postulated to induce a transient relaxation of tense muscle fibers, facilitating enhanced perfusion and adenosine triphosphate (ATP) replenishment. This process, in turn, facilitates the release of actin-myosin chains, leading to elongation of the muscle fiber and the subsequent elimination of metabolite waste. These cumulative effects contribute to the disruption of the pain-tension cycle. Indications for TPI encompass the identification of palpable trigger points that elicit a referred pain pattern and a twitch response [53].

The hallmark of needling therapy is the production of a local twitch response in the targeted muscle. It is theorized that the needle mechanically disrupts and stops the dysfunctional activity of the motor end plate of the skeletal muscle motor neuron [49]. Different techniques exist, including a direct method and a fanning approach injecting the trigger point with the utilization of ultrasonography for direct visualization and a palpatory based mode. However, there is no clear evidence that one method is superior to the other [54]. The use of ultrasound may be favorable to minimize complications from the procedure, such as vascular injury, pneumothorax, and injury to adjacent structures [53]. Fluoroscopy and ultrasound are the most commonly used imaging techniques to perform trigger point injections, in particular fluoroscopy is used for the piriformis muscle [1].

Ultrasound has been shown to provide higher accuracy in needle placement [55]. The administration of TPI is contraindicated under certain conditions such as the presence of infection, whether systemic or localized, coagulopathy, and patient refusal or highly anxious. TPIs are categorized as a low-risk medical procedure and usually stopping anticoagulants may not be required, depending on the body region [1].

TPI involves anesthetics, steroids, or botulinum toxin use. Notably, the utilization of shorter-acting and lower concentration anesthetics, such as lidocaine, has demonstrated diminished myotoxicity compared to long-acting anesthetics like bupivacaine, and a reduced injection-related discomfort in comparison to higher concentrations. Hyperirritable loci of trigger points in myofascial pain syndrome are known to contain vasoactive mediators, algogenic neurotransmitters and inflammatory mediators. While these are acknowledged to play a contributory role in myofascial pain syndrome, the efficacy of steroids in the context of TPI remains circumscribed [3,49]. Multiple studies collectively fail to furnish substantial evidence indicating that one injection modalities confer superior pain relief compared to dry needling as an isolated intervention. In addition, overall, there is no conclusive evidence that one TPI technique is more effective than another [3,49]. Outcome data pertaining to TPI for myofascial pain syndrome have demonstrated good efficacy with low safety profile through the administration of local anesthetics and corticosteroids. However, there is no substantiated evidence indicating the superiority of any specific injectate over others in achieving therapeutic benefits [1].

7.4. Botulinum toxin injection

Botulinum neurotoxin (BoNT) was first described in the 1800's when German physicians noted symptoms in people who consumed improperly preserved foods that grew *clostridium botulinum*. Subsequently, the neuroactive substance created by this bacterium was isolated and determined to act on the parasympathetic and motor nerve endings by way of calcium dependent inhibition of acetylcholine release at motor end plates [56]. Two notable serotypes were identified (BoNT-A and BoNT-B) each with their unique site-specific mechanism of action in disrupting vesicular fusion and neurotransmitter release at the presynaptic nerve terminal. Both BoNT-A and BoNT-B have been utilized for treatment of muscular disease with some success [57]. BoNT-A cleaves the synaptosomal-associated protein of 25 kd (SNAP 25) whereas BoNT-B cleaves the vesicular associated membrane protein (VAMP) [57]. This binding is reversible with a duration of action ranging from 3 to 4 months [58].

There are few studies evaluating the use of BoNT for treatment of myofascial pain and the information regarding its efficacy is mixed. Several studies including those published by Qerama et al., Kwanchuay et al., and Dessie et al. compared injections of BoNT-A to normal saline TPI in patients with shoulder, upper trapezius and pelvic myofascial pain, respectively. These studies found that there was no statistically significant increase in efficacy when patients underwent BoNT-A compared to saline TPI at the assessed endpoints; though both were effective in treating myofascial pain [59–61]. Interestingly, the study by Kwanchuay et al. did find that the pressure pain threshold was increased by BoNT-A compared to saline [60]. Aside from these studies, several others exist that have found similar findings that BoNT-A was effective but not superior to other injectates for TPI for lumbar myofascial pain and cervical myofascial pain [24].

Other studies have found that BoNT-A may have greater efficacy compared to their study comparator. Gobel et al. evaluated patients with upper back myofascial pain treated with BoNT-A compared to saline TPI. Their study found that BoNT-A provided statistically significant improvement in pain at 5 weeks with greater periods of fewer days with pain out to 12 weeks after initial treatment compared to saline TPI [62]. A study by Kamanil et al. evaluated BoNT-A TPI to lidocaine TPI, and dry needling to treat cervical and perscapular myofascial pain found that TPI was more effective than dry needling though BoNT-A did not confer greater relief than lidocaine based TPI though BoNT-A based TPI may be considered for medically resistant cases of myofascial pain [63].

Though well designed are few, studies available do indicate that BoNT-A based TPI may provide equivalent in efficacy to other TPI injectates. In medically refractory cases where patients have failed non BoNT-A TPI, consideration could be placed in trying BoNT-A based injections due to the potential conferred benefit of high pressure pain thresholds.

7.5. Behavioral interventions

Psychological based therapies and mindfulness techniques are the main means of behavioral management of chronic musculoskeletal pain [64]. Cognitive behavioral therapy (CBT) has been the stalwart and most well studied psychological therapy for treatment of chronic pain. However, few formalized studies exist evaluating the efficacy of behavioral interventions for myofascial pain. Those in existence evaluated the efficacy of group CBT in reducing perceived pain in patients with masticatory myofascial pain. The study by Bogart et al. found that some statistical improvement in pain intensity, frequency, and duration after CBT treatment compared to prior [65]. However, more studies exist evaluating the efficacy of behavioral interventions in all cause musculoskeletal pain.

The Agency for Healthcare Research and Quality found that psychological therapies were effective in providing small improvements in chronic low back pain [66]. A study by Cherkin et al. performed a randomized controlled trial comparing usual conservative care to CBT and mindfulness-based stress reduction (MBSR) in the treatment of patients with all cause chronic low back pain [67]. All 3 study groups received routine usual care for management of their low back pain but the intervention group (CBT and MBSR) received the intervention treatment while the usual care group was provided $50 to pursue whatever means of additional treatment they desired. Their study found that CBT and MBSR provided improvement in pain and debility in the study patients compared to usual care group at their 26 week follow up.

The American College of Physicians has published a clinical practice guideline evaluating the role of psychological therapy for low back pain. Their analysis found that overall, there is low quality evidence regarding progressive relaxation therapy and CBT's ability to reduce pain intensity and improve functioning. However, moderate level evidence was found supporting MBSR's efficacy in improving pain and functioning in patients with chronic pain [68]. Further studies are needed to fully evaluate the efficacy of behavioral therapies specific for myofascial pain however there is promise given it's modest effects in treating all cause back pain.

8. Conclusion

The challenging nature of MPS with its complex interaction of underlying biomechanical, neurologic, and psychological factors requires an astute, well-trained clinician for early diagnosis and effective treatment. Current data suggest that patients with MPS tend to do worse than patients with discreet musculoskeletal pathology such as disk herniation [69]. MPS can often be refractory to treatment. If the patient is unable to participate in an active functional rehabilitation program due to the limitations of pain, then a comprehensive approach should be considered. A multidisciplinary approach may include psychological counseling for coping strategies, relaxation techniques and biofeedback, cognitive behavioral therapy, and complementary and alternative medicine in addition to standard medical evaluation and management. When a comprehensive program is employed early in the disease course, functional improvement in measures of decreased sick time, improved coping skills, and increased life satisfaction can be demonstrated in patients with whiplash trauma and other MPSs [70]. Should pain persist, it is important to assess the potential contribution of coexisting musculoskeletal or neurologic pathology to MPS. The meticulous clinician should seek to identify and eliminate any underlying source of pain generation. However, despite an abundance of clinical experience and successful outcomes, we need better-designed, short- and long-term outcome studies on myofascial pain to assess the efficacy and efficiency of traditional and emerging therapies.

Practice points

- Myofascial pain syndrome is a type of regional soft tissue pain syndrome involving muscles of the trunk and extremities.
- Although myofascial pain may generalized, it remains distinct from fibromyalgia.
- Hyperirritable loci of trigger points have been found to contain vasoactive mediators, algogenic neurotransmitters, and inflammatory mediators.
- Excessive acetylcholine leakage has been hypothesized to contribute to dysfunctional motor end plates, creating the sustained muscle contraction responsible for taut bands.
- The clinical manifestation of abnormal electrical activity in the trigger point is a local twitch response, thought to be mediated by a segmental spinal reflex. Snapping palpation or needling the trigger point causes a brisk muscle contraction in the taut band.
- Diagnostic findings in the physical examination include observation of trigger points in an affected muscle, referral of pain to a zone of reference, and reproduction of the patient's usual pain.
- Early diagnosis and treatment with a comprehensive multimodal approach is optimal.

Research points

- While there is evidence of the efficacy of trigger point injection for myofascial pain, there is mixed evidence of the advantage of one injection technique over another, or the injection of any substance versus dry needling. Further large-scale, standardized, and randomized controlled trials are needed to investigate further.
- Injection of botulinum toxin is an emerging therapy that may be considered in refractory cases of myofascial pain, although evidence of its efficacy is limited at present. Future well-designed and rigorous randomized controlled trials are recommended.

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CRediT authorship contribution statement

Andrea L. Chadwick contributed to the conceptualization. All authors performed literature searches and data analysis. All authors were involved in drafting, writing, and revising the manuscript. All authors read and approved the final manuscript.

Declaration of competing interest

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